Apartment & Roommate Finder

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1. Problem Definition

Our Apartment Roommate Finder system tackles a challenge university students face when finding reliable off-campus housing and the perfect roommate in a structured, scalable, and reliable way. Current approaches are more fragmented across social media, group chats, spreadsheets, and word of mouth, making information inconsistent and hard to work with. Listings can be outdated and incomplete, with unclear roommate expectations. This makes it difficult to verify all sorts of information, like whether a listing or user is valid and whether roommates are compatible.

From a data management perspective, these issues exist due to poor handling of the underlying data. Students and providers need to manage connected information like users, listings, preferences, messages, matches, saved items, reports, and messages in a way that is well organized and enforces consistency. Without a centralized database model, it is difficult to support proper and efficient filtering, compatibility scoring, and reliable auditing/moderation.

The scope of this system is to centralize apartment listings, roommate profiles, and preference data into a single database-driven platform that supports structured searching and matching. The proposed solution resolves these challenges by defining clear entities, relationships, and integrity constraints that reflect real workflows, enabling accurate, efficient queries and minimizing mismatches, misinformation, and wasted effort.

2. System Users and Privileges

Admin

- Manage all system records (users, student profiles, roommate profiles, providers, listings, saved items, matches, and conversations)
- Moderate the platform by reviewing/removing fraudulent or inappropriate listings/profiles and handling reported issues
- Enforce verification/trust rules and resolve account or data integrity issues (e.g., deactivations, duplicates)

Student User (Registered User with StudentProfile and optionally a RoommateProfile)

- Create and maintain account and student profile information
- Create/update roommate profile and preferences to participate in roommate matching
- Browse and filter apartment listings and roommate profiles
- Save/bookmark listings or roommate profiles for later reference (SavedItem)
- Start/join conversations and send messages to providers or other students (Conversation, Message)

Apartment Provider (Registered User with ApartmentProvider)

- Create and maintain provider profile information
- Create, update, and archive apartment listings and manage listing images (ApartmentListing, ListingImage)
- Communicate with interested students through the messaging system (Conversation, Message)

3. Entities and Their Descriptions

User
Represents a registered account in the system. Users may act as students and/or apartment providers and serve as the owner/creator for related records

StudentProfile
Represents student-specific identity information linked to a User. Used to model students as a distinct role and support student-facing features.

ApartmentProvider
Represents a provider role linked to a User account. Used to model property owners/managers who create and manage apartment listings.

ApartmentListing
Represents an apartment listing posted by an ApartmentProvider. Used to store and manage housing options available to students.

ListingImage
Represents an image resource associated with an ApartmentListing. Used to support multiple images per listing and maintain display ordering.

RoommateProfile
Represents a student's roommate seeking a profile linked to a User. Used to support roommate discovery and compatibility matching.

Match
Represents a compatibility result between two RoommateProfiles. Used to store match outcomes and support ranking/recommendations.

Conversation
Represents a messaging thread in the system. Used to group communication related to a listing or roommate connection.

ConversationParticipant
Represents the association between a User and a Conversation. Used to model conversation membership (one-to-one or group) and participant tracking.

Message
Represents a single message posted within a Conversation. Used to store user-to-user communication history.

SavedItem
Represents a bookmark created by a User for later reference. Used to save either apartment listings or roommate profiles for quick access.

4. Relationships and Their Descriptions

User: StudentProfile (1 : 0..1)
A User may have at most one StudentProfile. This relationship represents that only users who are students have a student-specific profile.

User: ApartmentProvider (1 : 0..1)
A User may have at most one ApartmentProvider profile. This relationship represents users who act as property managers/owners within the system.

User: RoommateProfile (1 : 0..1)
A User may have at most one RoommateProfile. This relationship represents students who opt into roommate matching by creating a roommate-seeking profile.

ApartmentProvider: ApartmentListing (1 : many)
An ApartmentProvider can create and manage multiple ApartmentListings. Each listing is owned by exactly one provider.

ApartmentListing: ListingImage (1 : many)
An ApartmentListing can have multiple ListingImages. Each image belongs to exactly one listing, allowing listings to display photo galleries.

RoommateProfile: Match (many : many via Match table)
A Match represents a compatibility result between two RoommateProfiles. Each RoommateProfile can appear in many matches, and each match links exactly two roommate profiles.

ApartmentListing: Conversation (1 : many, optional on Conversation)
A Conversation may be associated with an ApartmentListing (e.g., a student messaging a provider about a listing). A single listing may have multiple conversations with different users.

RoommateProfile: Conversation (1 : many, optional on Conversation)
A Conversation may be associated with a RoommateProfile (e.g., students messaging about roommate interest). A single roommate profile may have multiple conversations.

Conversation: ConversationParticipant (1 : many)
A Conversation can have multiple ConversationParticipants. This relationship tracks which users are part of a given conversation.

User: ConversationParticipant (1 : many)
A User can appear in many ConversationParticipant records, allowing a user to participate in multiple conversations (supports many-to-many between User and Conversation).

Conversation: Message (1 : many)
A Conversation contains many Messages. Each message belongs to exactly one conversation.

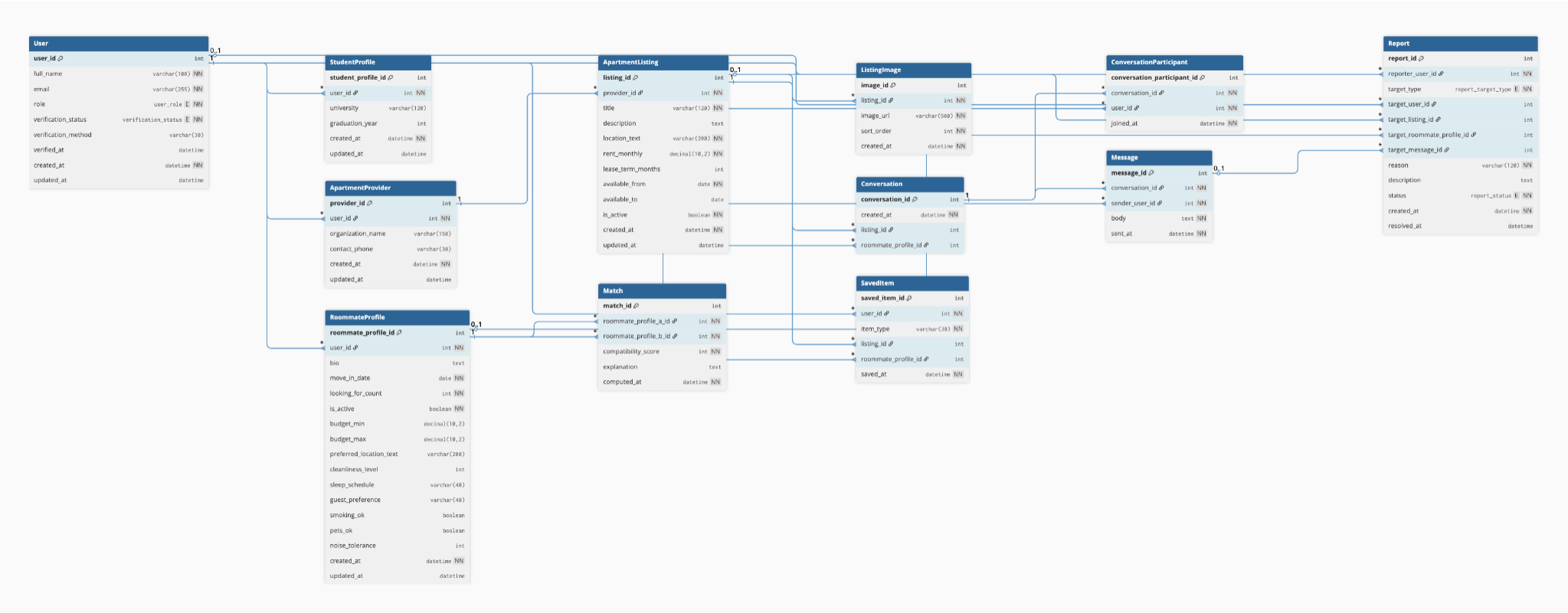
User: Message (1 : many)
A User can send many Messages. Each message is sent by exactly one user (the sender), enabling auditability and messaging history.

User: SavedItem (1 : many)
A User can create many SavedItems. This relationship supports bookmarking of listings or roommate profiles.

ApartmentListing: SavedItem (1 : many, optional on SavedItem)
A SavedItem may reference an ApartmentListing, enabling users to save listings for later viewing. A listing can be saved by many users.

RoommateProfile: SavedItem (1 : many, optional on SavedItem)
A SavedItem may reference a RoommateProfile, enabling users to save roommate profiles. A roommate profile can be saved by many users.

5. Entity Relationship Diagram (ERD)



6. Roles of Project Members

Daniel Carrillo

Role:

- Contribute to database schema design and refinement, including entity relationships and constraints
- Assist in defining system requirements and translating them into technical specifications
- Design and document value-added system features with a focus on scalability and data integrity
- Support backend development by validating queries, constraints, and data workflows
- Contribute to testing and verification to ensure correctness and consistency across system components
- **Worked on System Users and Privileges - Phase 2**

Hursh Soni

Role:

- Assist with DB design and implementation (data models, relationships, etc.)
- Work on frontend design & development
- Assist with backend development
- Work on integration testing
- Project documentation through the semester
- **Worked on ERD - Phase 2**
- **Documentation - Phase 2**

Anthony Osei

Role:

- Work on backend development
- Work on project phase documentations
- Assist with database design and implementation
- **Worked on problem definition and helped with system users and privileges - Phase 2**

Vivan Verma

Role:

- Work with Anthony to ensure a well-developed backend
- Work with Hursh to also develop a frontend
- Testing
- **Worked on problem definition entities and their description and relationships - Phase 2**

Generative AI

Role:

- Help with debugging/identifying issues
- Assist with system architecture and design
- Help understand the project scope and provide ideas
- **Clean up tasks, help with documentation, and ERD - Phase 2**
- **Feedback - Phase 2**